

■ PHYSICS (Class 11 Core JEE Topics)

1. Units & Dimensions

Subtopics:

1. Physical Quantities
 2. Fundamental & Derived Units
 3. Dimensional Formulae
 4. Dimensional Analysis
 5. Checking correctness of equations
 6. Conversion of units
 7. Significant figures & Errors (very important for JEE)
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2. Kinematics

Subtopics:

Motion in Straight Line

1. Distance, Displacement
2. Speed, Velocity
3. Acceleration
4. Graphs ($v-t$, $x-t$, $a-t$)
5. Equations of Motion
6. Relative Motion

Motion in a Plane

7. Vectors
 8. Projectile Motion
 9. Relative Velocity in 2D
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3. Newton's Laws of Motion

Subtopics:

1. Inertia, Force

2. Free Body Diagram (FBD)
 3. Newton's 1st, 2nd, 3rd Law
 4. Normal Reaction, Tension
 5. Pseudo Force (important for JEE)
 6. Friction
 - Static, Kinetic
 - Rolling friction
 7. Circular Motion
 - Centripetal Force
 - Banking of roads
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4. Work, Energy & Power

Subtopics:

1. Work Done by Constant & Variable Force
 2. Kinetic & Potential Energy
 3. Work–Energy Theorem
 4. Power
 5. Conservation of Energy
 6. Conservative & Non-conservative Forces
 7. Springs (Elastic Potential Energy)
 8. Collisions (Elastic & Inelastic)
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CHEMISTRY (Class 11 Core JEE Topics)

1. Mole Concept & Stoichiometry

Subtopics:

1. Mole, Molar Mass
 2. Empirical & Molecular Formula
 3. Limiting Reagent
 4. Percentage Composition
 5. Stoichiometric Calculations
 6. Equivalent Weight
 7. Molarity, Molality, Normality (basics)
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2. Atomic Structure

Subtopics:

1. Dual nature of matter
 2. Photoelectric effect basics
 3. Thomson, Rutherford Models
 4. Bohr's Model & Limitations
 5. Quantum Numbers
 6. Orbitals & Shapes
 7. de Broglie wavelength
 8. Heisenberg Uncertainty Principle
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3. Periodic Table & Periodicity

Subtopics:

1. Modern Periodic Law
 2. Trends:
 - Atomic radius
 - Ionization enthalpy
 - Electronegativity
 - Electron gain enthalpy
 - Metallic/Nonmetallic nature
 3. s, p, d, f blocks
 4. Anomalies (Be, B, O, F, etc.)
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4. Chemical Bonding

Subtopics:

1. Ionic Bond
 2. Covalent Bond
 3. Octet Rule
 4. VSEPR Theory (Shapes of molecules)
 5. Hybridization (sp , sp^2 , sp^3 , sp^3d , sp^3d^2)
 6. Molecular Orbital Theory (Basics)
 7. Polarity & Dipole Moment
 8. Hydrogen Bonding
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1. Sets, Relations & Functions (11th part)

Subtopics:

1. Sets (Union, Intersection, Complement)
 2. Venn Diagrams
 3. Cartesian Product
 4. Relations (Reflexive, Symmetric, Transitive)
 5. Functions (Domain, Range)
 6. Types of Functions
 7. Graph basics
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2. Quadratic Equations

Subtopics:

1. Standard Form
 2. Nature of Roots
 3. Discriminant
 4. Theory of Equations
 5. Quadratic Inequalities
 6. Graph of Quadratic Function
 7. Maximum/Minimum of Quadratic (Parabola concept)
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3. Trigonometry

Subtopics:

1. Trigonometric Ratios & Identities
 2. Trigonometric Equations
 3. Heights & Distances
 4. General Solutions
 5. Graphs of Trigonometric Functions
 6. Formulae (Very Important):
 - C&D formulas
 - Multiple & Sub-multiple angle
 - Product-to-sum, sum-to-product
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WEEK 1

Physics: Units & Dimensions

Chemistry: Mole Concept

Maths: Sets & Functions

WEEK 2

Physics: Kinematics

Chemistry: Atomic Structure

Maths: Quadratic Equations

WEEK 3

Physics: Newton's Laws

Chemistry: Periodic Table

Maths: Trigonometry (Basics + Identities)

WEEK 4

Physics: Work, Power, Energy

Chemistry: Chemical Bonding

Maths: Trigonometry (Advanced – equations + graphs)

Mixed PYQs + JEE-foundation questions